

IIT Jammu

Development of Autonomous Electric Vehicle for IIT Jammu Campus

Team Members :

- Nageshwar Kumar
- Saurav Kumar



Dr. Nalin Kumar Sharma

Problem Statement

- Current EVs lack autonomous driving capability required for intelligent transportation.
- Safe navigation in **structured** (lanes, signals) and **unstructured** (mixed traffic, Indian conditions) environments remains a major challenge.
- Need to integrate **sensors (LiDAR, cameras, IMU, GPS)**, **path planning**, and **control algorithms** to enable self-driving.
- Objective this semester: Implement and test core autonomous driving modules (**perception, decision-making, and control**) on our EV platform.

Objectives

1. Lane Detection - Necessity

- Enables real-time lateral position control by detecting lane boundaries through computer vision for autonomous path following
 - Facilitates navigation decision-making by identifying lane markings, merge points, and intersection geometries on campus roads
 - Provides visual feedback for trajectory correction algorithms to prevent lane departure and maintain safe vehicle positioning

Objectives

1. Lane Detection - Solution

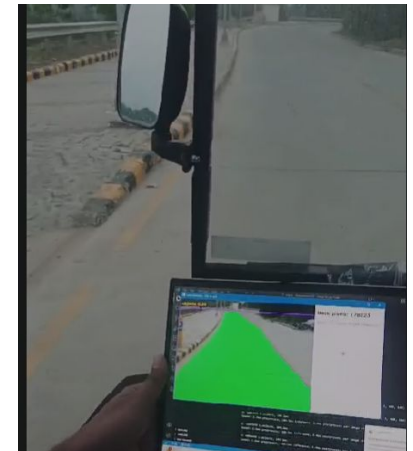
- *Completed:* Implemented lane detection using a YOLO segmentation model trained on our own annotated dataset. Custom lane masks and boundary labels were created from scratch to improve accuracy in our driving environment.
- *Validated* lane detection shows perfect performance in daylight, minor flickering in evening conditions, and unreliable results at night, requiring LiDAR integration or alternative night-vision methods.
- *Future Plan:* Explore night-vision methods and develop junction-decision algorithms for fully autonomous navigation.



Fig: Annotated Images



Fig: Lane Detected



Objectives

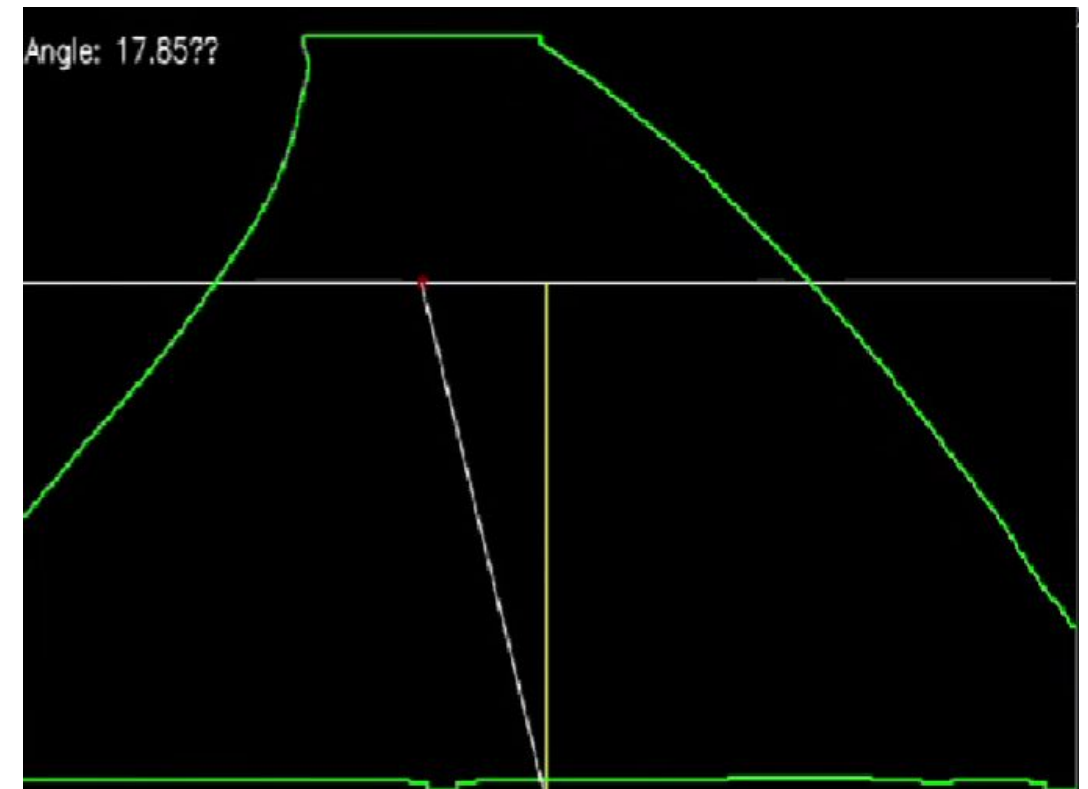
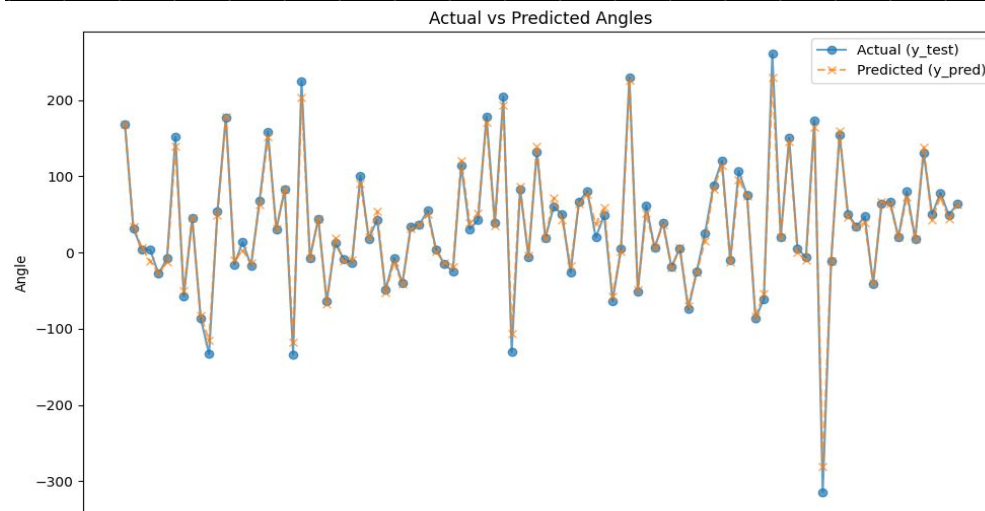
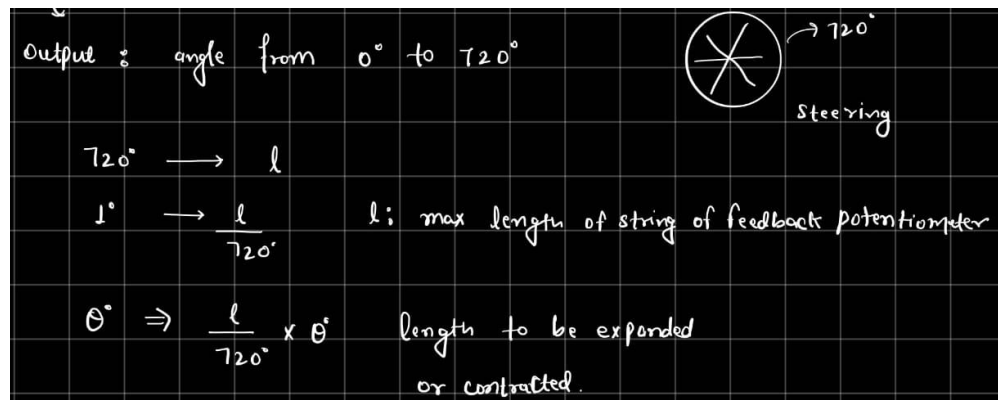
2. Steering Angle Prediction - Necessity

- Enables dynamic path following by predicting optimal steering angles in real-time based on road geometry and vehicle state.
- Serves as critical input for the steer-by-wire system, ensuring accurate actuation aligned with aid of navigation commands.
- Enhances vehicle stability by minimizing abrupt steering corrections during autonomous operation.

Objectives

2. Steering Angle Prediction - Solution

- Extracted lane masks using the YOLO segmentation model, converting them into polygon contours for geometric processing.



Objectives

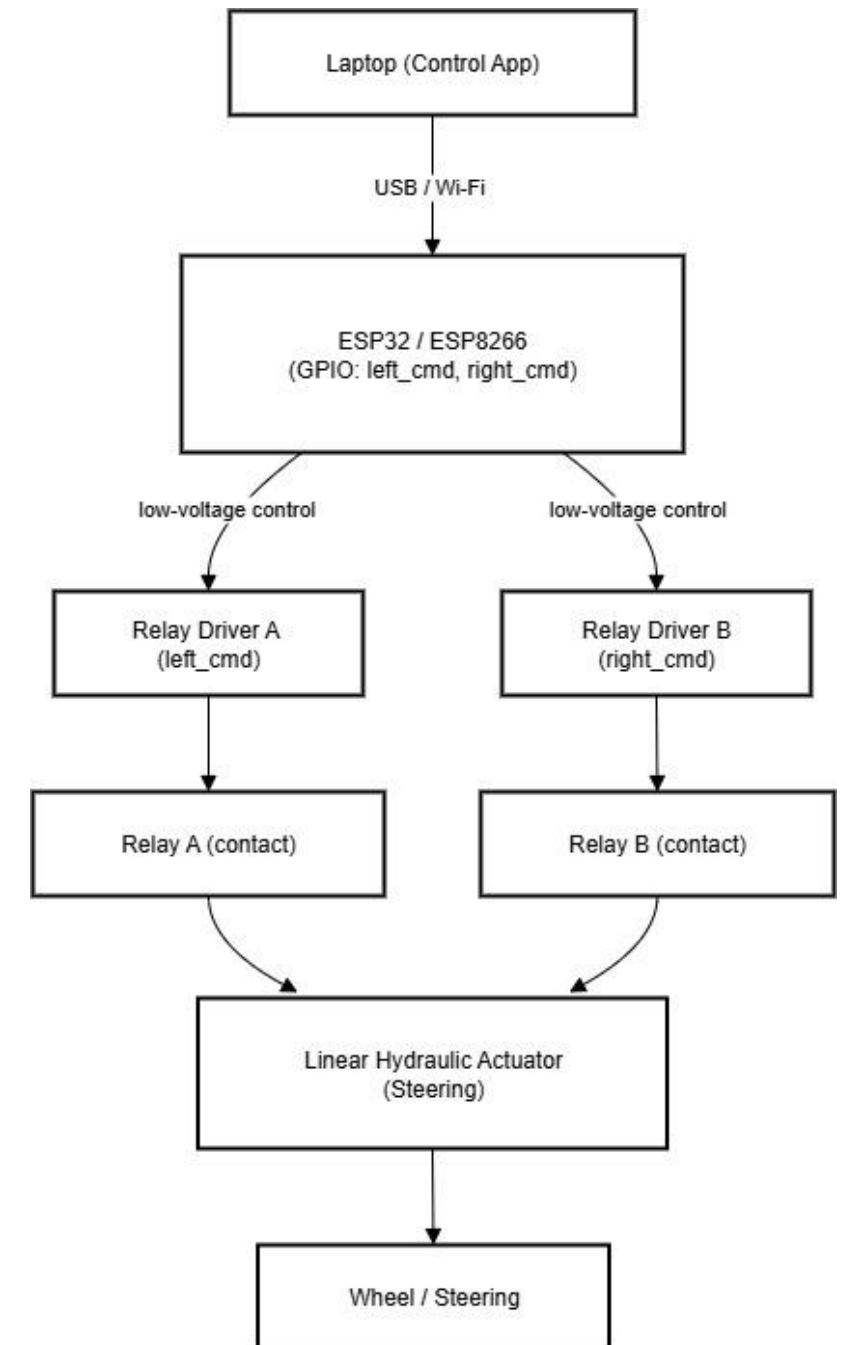
3. Steering Control Module Development - Necessity

- Acts as the basis for real-time interface that converts autonomous planning outputs (**steering, braking, throttling**) into low-level actuator commands for hydraulic throttle/brake systems.
- Enables Integration of sensor feedback (potentiometer data) for closed-loop control, improving precision in speed modulation and braking force application.

Objectives

3. Steering Control Module Development - Solution

- Implemented **steer-by-wire** system using **hydraulics** for throttle and brake.
- *Problem:* Absence of feedback loop leads to compounding steering errors

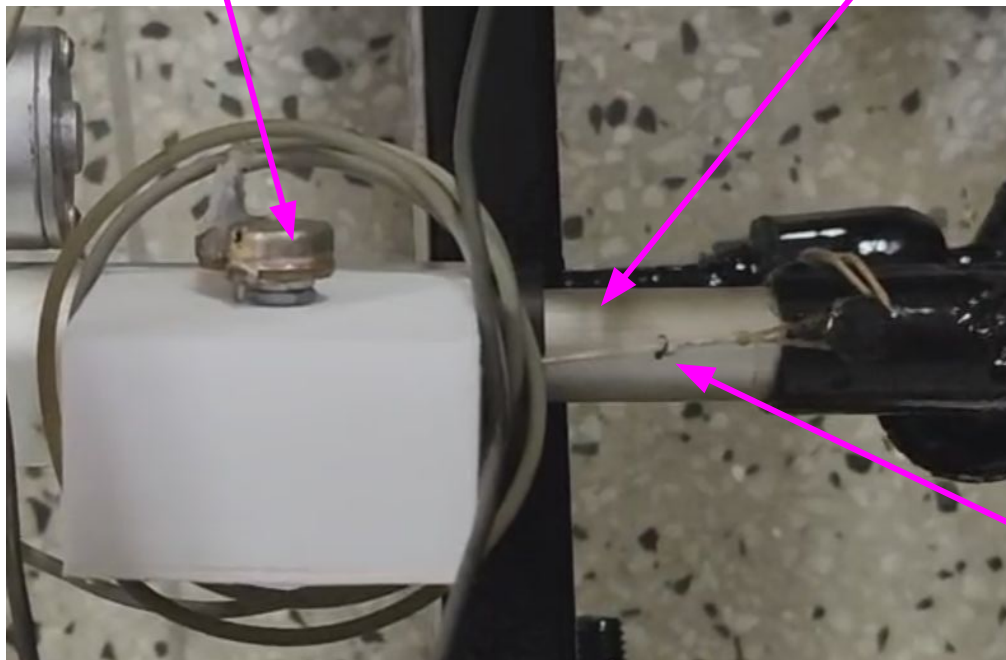


Objectives

3. Steering Control Module Development

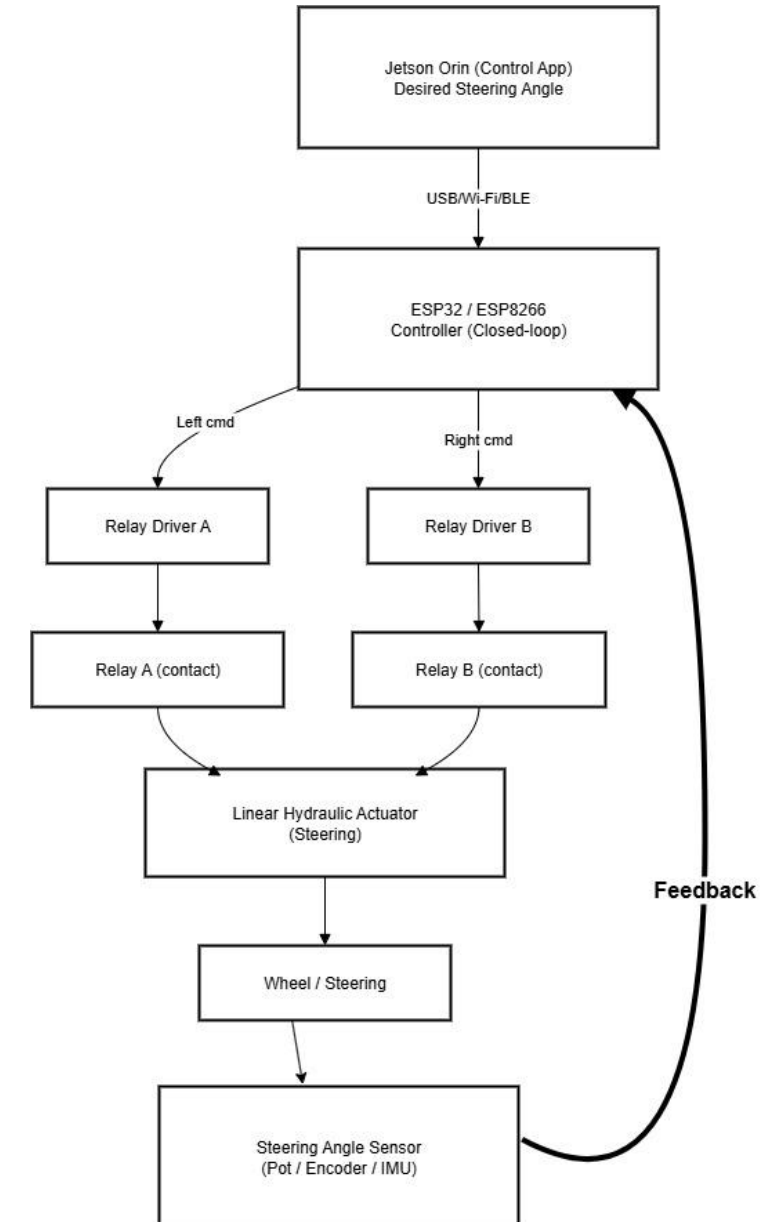
Rotary Potentiometer

Hydraulic limb



String

Fig: String Potentiometer



Objectives

Potentiometer-based feedback

- A potentiometer is a three terminal resistive divider circuit.

$$V_{out}(\theta) = V_{cc} \cdot \frac{\theta}{\theta_{max}}$$

V_{cc} = supply voltage across potentiometer.

θ = angular displacement of the potentiometer shaft (caused by actuator thread tension)

θ_{max} = maximum mechanical rotation range of potentiometer

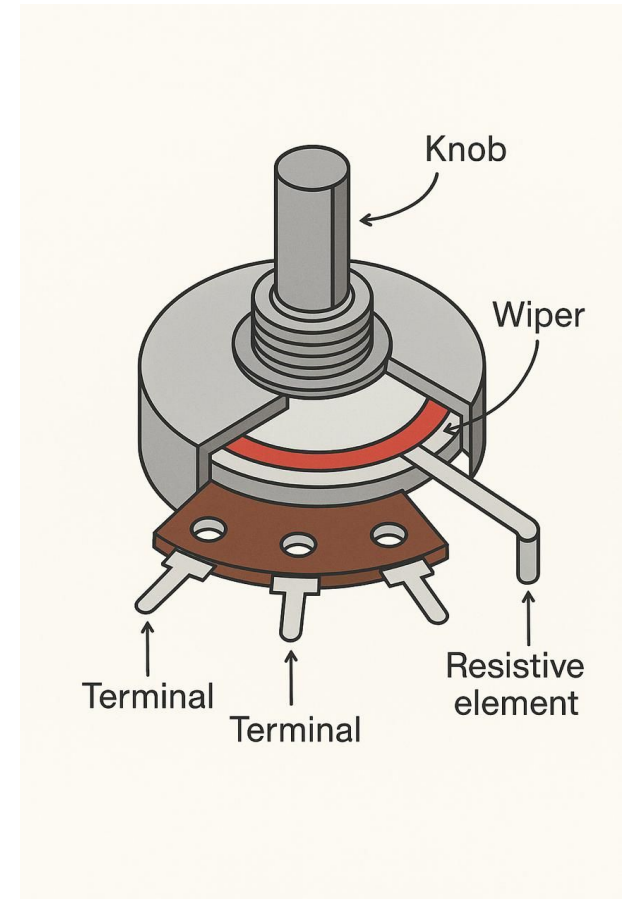
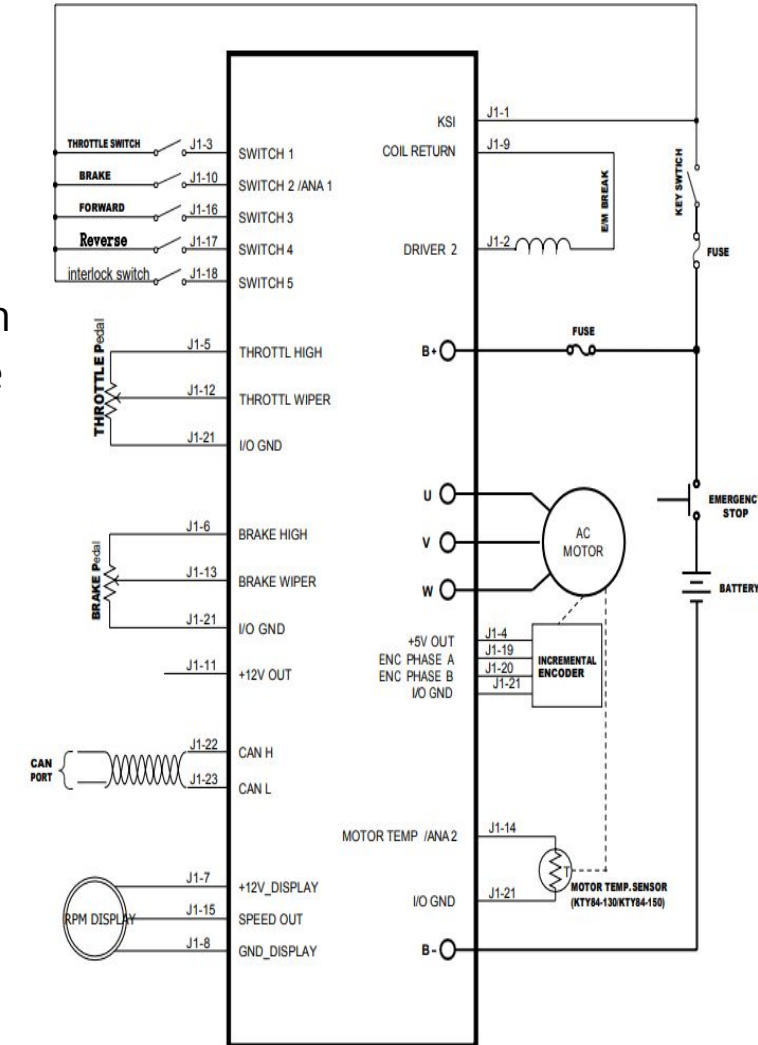


Fig: Rotary Potentiometer

Objectives

Vehicle's Control Module

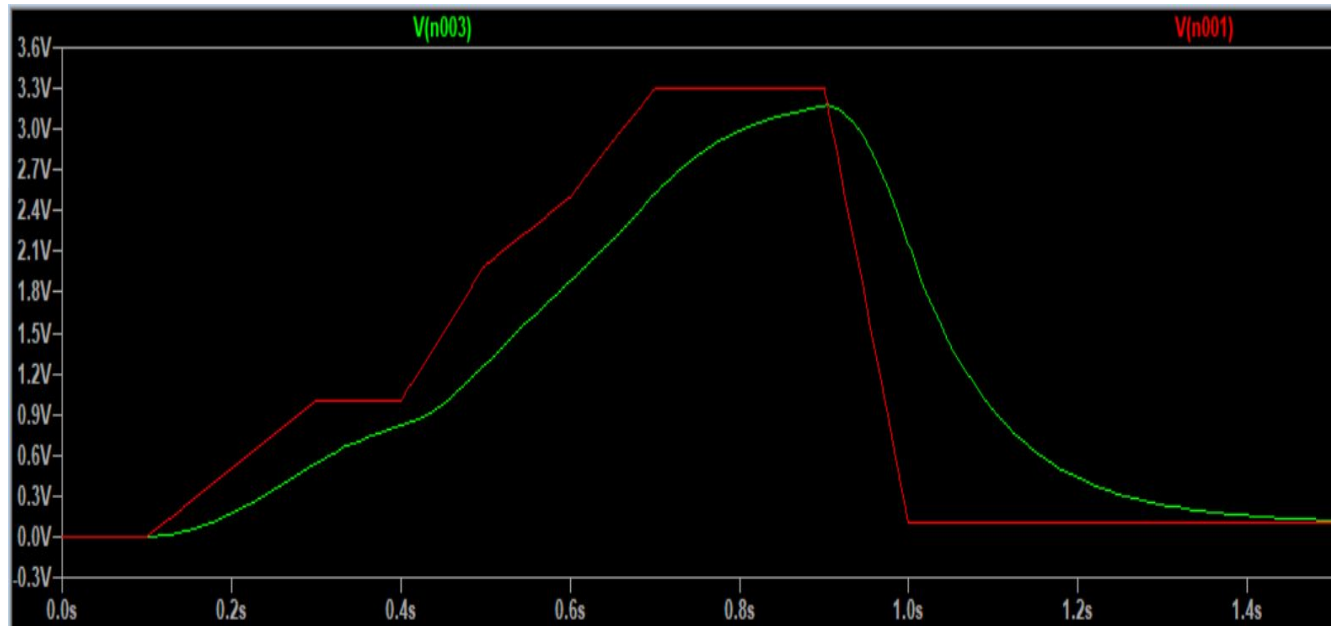
- Throttle Pins** (J1-5: *THROTTLE HIGH*, J1-12: *THROTTLE WIPER*, J1-21: *I/O GND*): These three pins form a standard **0–6V throttle input** where the microcontroller outputs a smooth analog voltage between *THROTTLE HIGH* and *I/O GND*, and the controller reads the effective command through the **THROTTLE WIPER** line to regulate vehicle acceleration.
- Brake Pins** (J1-6: *BRAKE HIGH*, J1-13: *BRAKE WIPER*): Similar to the throttle interface, these pins provide a **0–6V analog brake signal**, where the **BRAKE WIPER** voltage determines braking intensity; applying a high brake command automatically overrides throttle output for safe stopping.



Objectives

Throttle - Jerk Prevention

- Spice Simulation



Input: PWM in

Output: Filtered PWM

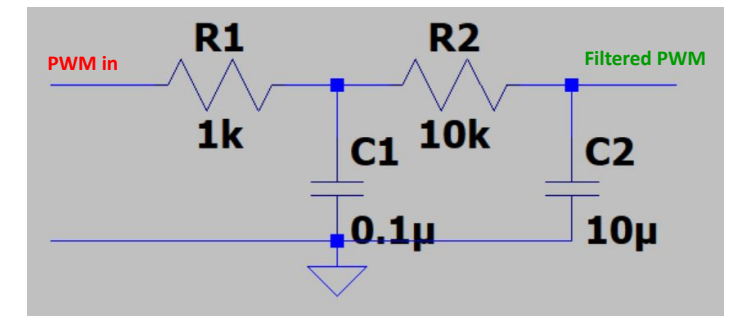
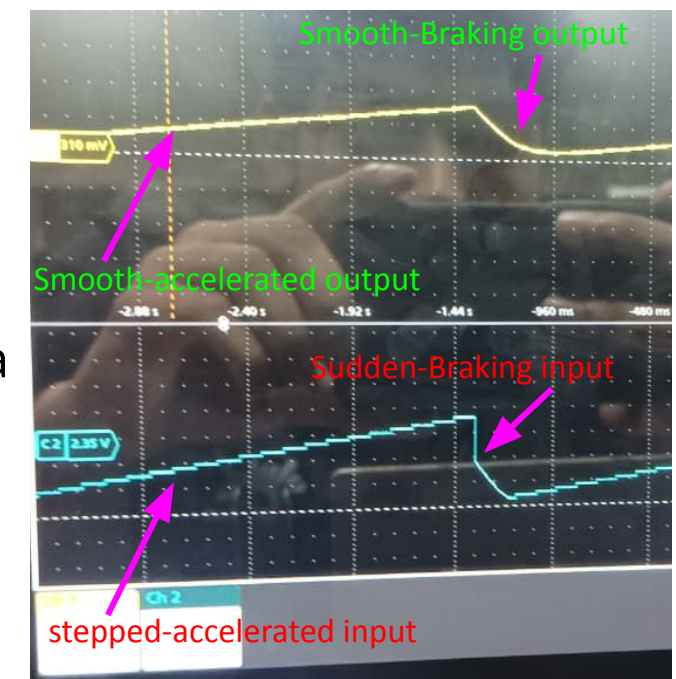
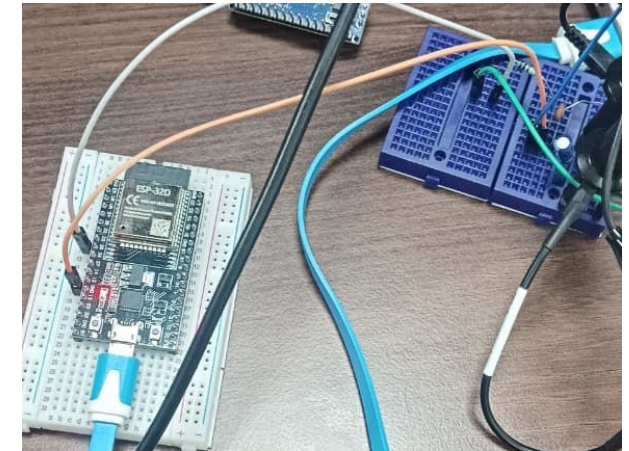


Fig: RC Low pass filter

Objectives

Throttle - Jerk Prevention

- The EV motor controller interprets a 0–5V analog throttle signal, and this voltage is generated by converting the ESP's PWM output into a stable analog level using an RC low-pass filter.
- The filtered throttle voltage is fed directly into the controller's Throttle Input pin, enabling the microcontroller to regulate acceleration and maintain a fixed cruise-speed ratio.



Objectives

Brake Control

- The controller receives the brake signal through **BRAKE WIPER(J1-6)** which is in range of **BRAKE HIGH (J1-13)** and **GND**, where the voltage on the wiper pin (0–5V) tells the system **how much braking force** to apply.
- In autonomous mode, the microcontroller sends this brake voltage directly, and any brake command **instantly overrides the throttle**, ensuring the vehicle slows down safely.
- Status : planned

Objectives

Obstacle Avoidance

- Measurement Range: 20 cm to 600 cm (0.2–6 m)
- Best Accuracy Range: 30 cm to 200 cm
- We ordered SR04M ultrasonic sensors intended for side-wise obstacle detection, but are currently using one as a front proof-of-concept sensor.
- The sensor's distance reading is monitored by the controller; when an obstacle is detected within the safety threshold, the system immediately sets throttle voltage to zero and raises braking voltage to a high level to perform an emergency stop.

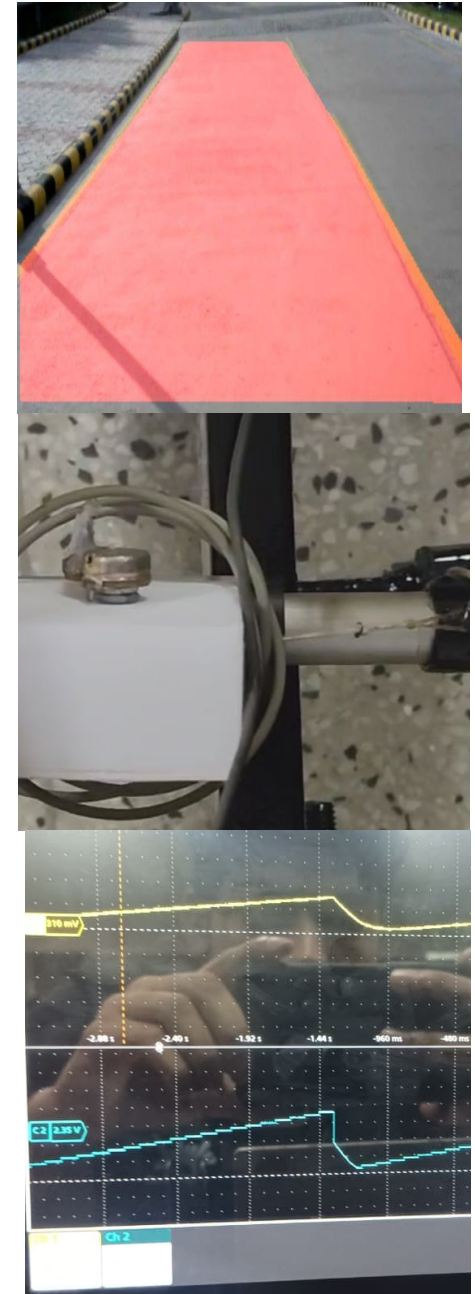


Fig: SR04M Ultrasonic Sensor

Conclusion

Sensor-controller integration

- The lane-detection module identifies road boundaries and computes the required steering direction.
- The feedback potentiometer reports the real-time steering angle for closed-loop control.
- The vehicle auto-accelerates to a predefined speed by regulating the throttle voltage to a set ratio. (path-planning modules planned for future upgrades)
- The obstacle-avoidance system cuts the throttle to zero and simultaneously applies braking voltage to ensure an immediate and safe stop.



- **Simulation in Gazebo**

- Set up the Gazebo simulator with the city road and urban environment loaded and functioning.
- Successfully spawned the Prius model in the simulation, verified via the Gazebo interface and model topics.
- Enabled keyboard (WASD) controls, allowing manual driving of the Prius within the environment.
- Activated the vehicle's virtual sensors in ROS, including multiple camera and ultrasonic sensors suitable for perception and lane detection work.
- Verified front camera ROS topic is active and ready for lane following or detection node development.



विद्यार्थनं सर्वधनं प्रधानम्



